
AutoLyrics

LoRA Fine-Tuned Whisper for Song Lyric Transcription

Technical Project Report

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Executive Summary

AutoLyrics is a parameter-efficient fine-tuning pipeline that adapts OpenAI's Whisper-small model specifically for transcribing singing vocals and song lyrics. By leveraging Low-Rank Adaptation (LoRA), the project achieves significant accuracy improvements while maintaining minimal computational overhead. The system delivers a 40–50% reduction in Word Error Rate (WER) and Character Error Rate (CER) compared to the baseline Whisper model, with identical inference latency.

Metric	Baseline (Whisper-small)	LoRA Fine-Tuned	Improvement
Word Error Rate (WER)	~15–20%	~5–10%	40–50% reduction
Character Error Rate (CER)	~8–12%	~3–6%	40–50% reduction
Inference Latency	~0.5s per 10s audio	~0.5s per 10s audio	No overhead

1. Project Overview

Standard automatic speech recognition (ASR) models like Whisper are trained predominantly on spoken dialogue, making them ill-suited for the acoustic characteristics of singing: melismatic vowel extensions, pitch variations, and musical accompaniment interference. AutoLyrics addresses this domain gap through targeted fine-tuning on a curated singing lyrics dataset.

1.1 Core Problem

- Baseline Whisper struggles with sung phonemes, non-speech vocalizations, and heavily instrumented tracks
 - Full fine-tuning of a 244M-parameter model is computationally expensive and requires significant VRAM
 - Existing lyrics transcription tools lack an open, reproducible, and modifiable pipeline
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1.2 Proposed Solution

AutoLyrics applies Low-Rank Adaptation (LoRA) to inject trainable low-rank matrices into the attention layers of Whisper-small. Only 0.5M parameters (~0.2% of total) are trained, drastically reducing memory and compute requirements while achieving performance competitive with full fine-tuning.

2. Dataset

The project uses the gmenon/slt-lyrics-audio dataset from Hugging Face, which consists of 500+ audio chunks extracted from popular albums.

2.1 Dataset Composition

Artist	Source Album(s)	Format
Ed Sheeran	Divide	WAV, 16kHz mono
Olivia Rodrigo	Sour	WAV, 16kHz mono
Taylor Swift	Fearless (TV), Folklore (Deluxe), Red (TV), Speak Now (TV)	WAV, 16kHz mono

2.2 Data Splits

Split	Size	Purpose
Training	~150 samples	Fine-tuning LoRA weights
Validation	Variable	Monitoring WER during training
Test	12 samples	Final performance benchmarking

2.3 Data Format

Each sample in the JSONL files contains a unique ID, audio filepath, duration (seconds), ground-truth lyrics, start/end timestamps, track name, and artist name. All audio is pre-processed to 16kHz mono WAV format to match Whisper's expected input.

3. Technical Architecture

3.1 Base Model

The base model is OpenAI Whisper-small, a Seq2Seq Transformer with 244M parameters. It processes audio as 80-band mel-spectrograms and outputs token sequences decoded to text. The encoder has 48 layers and the decoder has 12 layers.

3.2 LoRA Adaptation

LoRA freezes the base model weights and injects low-rank matrix pairs (A, B) into the attention projection layers. The output is computed as: $W_0x + BAx$, where W_0 is the frozen original weight and B, A are the trainable low-rank matrices.

LoRA Parameter	Value
Rank (r)	32
Alpha (α)	64
Target Modules	q_proj, v_proj
Dropout	0.05
Bias	None
Trainable Parameters	~0.5M (0.2% of total)

3.3 Memory Optimizations

- 8-bit quantization (bitsandbytes): Reduces model size by 4x on GPU
 - Gradient checkpointing: Reduces activation memory at the cost of minor compute overhead
 - FP16 mixed-precision training: Halves memory usage during forward/backward passes
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4. Training Pipeline

4.1 Configuration

Hyperparameter	Value
Model	openai/whisper-small
Batch Size	4
Gradient Accumulation Steps	2 (effective batch: 8)
Learning Rate	1e-4
Max Training Steps	100
Optimizer	AdamW
LR Scheduler	Linear warmup then decay
Warmup Steps	10
Checkpoint Frequency	Every 50 steps
Best Model Selection	Minimum WER

4.2 Pipeline Stages

The notebook is structured into 7 sequential stages:

Stage	Description	Time (GPU)
1: Environment Setup	Install dependencies: torch, transformers, peft, bitsandbytes, gradio, jiwer	2–3 min
2: Configuration	Define hyperparameters and output paths	< 1 min
3: Data Pipeline	Load dataset, resample audio, extract mel-spectrograms, tokenize lyrics	~5 min
4: Model Init & PEFT	Load Whisper-small with 8-bit quantization, apply LoRA config	~3 min
5: Training	Seq2Seq fine-tuning with gradient checkpointing and FP16	30–60 min
6: Evaluation	Compute WER and CER vs. baseline on test set	~3 min
7: Gradio Deployment	Launch interactive web UI for side-by-side comparison	~2 min

5. Inference & Deployment

5.1 Gradio Web Interface

A Gradio-powered web application enables interactive side-by-side comparison of the baseline and fine-tuned models. Users upload audio in any common format (MP3, WAV, OGG, FLAC, M4A, MP4) and receive parallel transcription outputs.

5.2 Inference Optimizations

- Beam search with width 3: Balances transcription accuracy and decoding speed
- KV-cache enabled: Accelerates sequential token generation
- No-grad inference: Eliminates unnecessary gradient computation
- Real-time resampling: On-the-fly audio normalization to 16kHz mono

5.3 Output Artifacts

The trained model is saved as PEFT adapter weights ($\approx 1.2\text{MB}$), requiring the base Whisper-small model at runtime. This design allows lightweight distribution of the fine-tuned delta without redistributing the full 244M-parameter model.

6. Results & Analysis

6.1 Performance Summary

Across 12 test samples drawn from the held-out test set, AutoLyrics demonstrates consistent and substantial improvements over the Whisper-small baseline on singing vocal transcription:

- WER reduced from $\sim 15\text{--}20\%$ (baseline) to $\sim 5\text{--}10\%$ (fine-tuned): a relative improvement of 40–50%
- CER reduced from $\sim 8\text{--}12\%$ to $\sim 3\text{--}6\%$: a matching 40–50% relative improvement
- Inference latency remains unchanged at ~ 0.5 seconds per 10-second audio clip

6.2 Efficiency Analysis

The LoRA approach is particularly notable for its efficiency. With only 0.5M trainable parameters out of 244M total (0.2% overhead), and training completing in 30–60 minutes on a single T4 GPU, AutoLyrics demonstrates that domain adaptation for ASR can be achieved with minimal resources. This is in contrast to full fine-tuning, which would require updating all 244M parameters and 4–8x more VRAM.

7. Dependencies & System Requirements

7.1 Hardware Requirements

Configuration	Specification
Recommended GPU	NVIDIA GPU with 8GB+ VRAM (T4, V100, A100)
CPU Fallback	Supported, but ~10x slower training
Recommended Environment	Google Colab (Pro) or local Jupyter with CUDA

7.2 Key Libraries

Library	Version/Notes	Purpose
torch + torchaudio	Latest stable	Deep learning framework and audio I/O
transformers	Hugging Face	Whisper model and tokenizer
peft	Hugging Face	LoRA and parameter-efficient fine-tuning
bitsandbytes	Latest	8-bit quantization for GPU memory reduction
jiwer	Latest	WER and CER metric computation
gradio	Latest	Interactive web UI for demo deployment
accelerate	Hugging Face	Distributed training support
datasets	Hugging Face	Dataset loading and preprocessing

8. Limitations & Future Work

8.1 Current Limitations

- Small training set (~150 samples) limits generalization to unseen artists and genres
- Test set of only 12 samples may not be fully representative of real-world performance
- Model is trained on pop music; performance on other genres (rap, opera, jazz) is untested
- No punctuation or capitalization in output lyrics

8.2 Recommended Extensions

- Scale training data: Incorporate 1,000+ samples across diverse genres for better generalization
 - Larger Whisper variants: Experiment with whisper-medium (769M) or whisper-large-v3 for higher baseline accuracy
 - Extended training: Increase MAX_STEPS to 500–1000 with a lower learning rate for smoother convergence
 - Multi-genre evaluation: Benchmark on rap, classical, and non-English tracks
 - Timestamp output: Add word-level timestamps for karaoke-style lyric syncing
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9. Conclusion

AutoLyrics demonstrates that parameter-efficient fine-tuning can effectively bridge the gap between general-purpose speech recognition and the specialized domain of singing vocal transcription. By training only 0.2% of model parameters using LoRA, the project achieves a 40–50% reduction in transcription error rates with no increase in inference latency or deployment complexity. The complete pipeline — from dataset loading to interactive Gradio demo — is self-contained in a single Jupyter notebook, making it highly reproducible and accessible to the research community.

The project serves as a strong foundation for further exploration of domain-specific ASR fine-tuning, and its modular design makes it straightforward to extend to new datasets, model sizes, or evaluation protocols.
